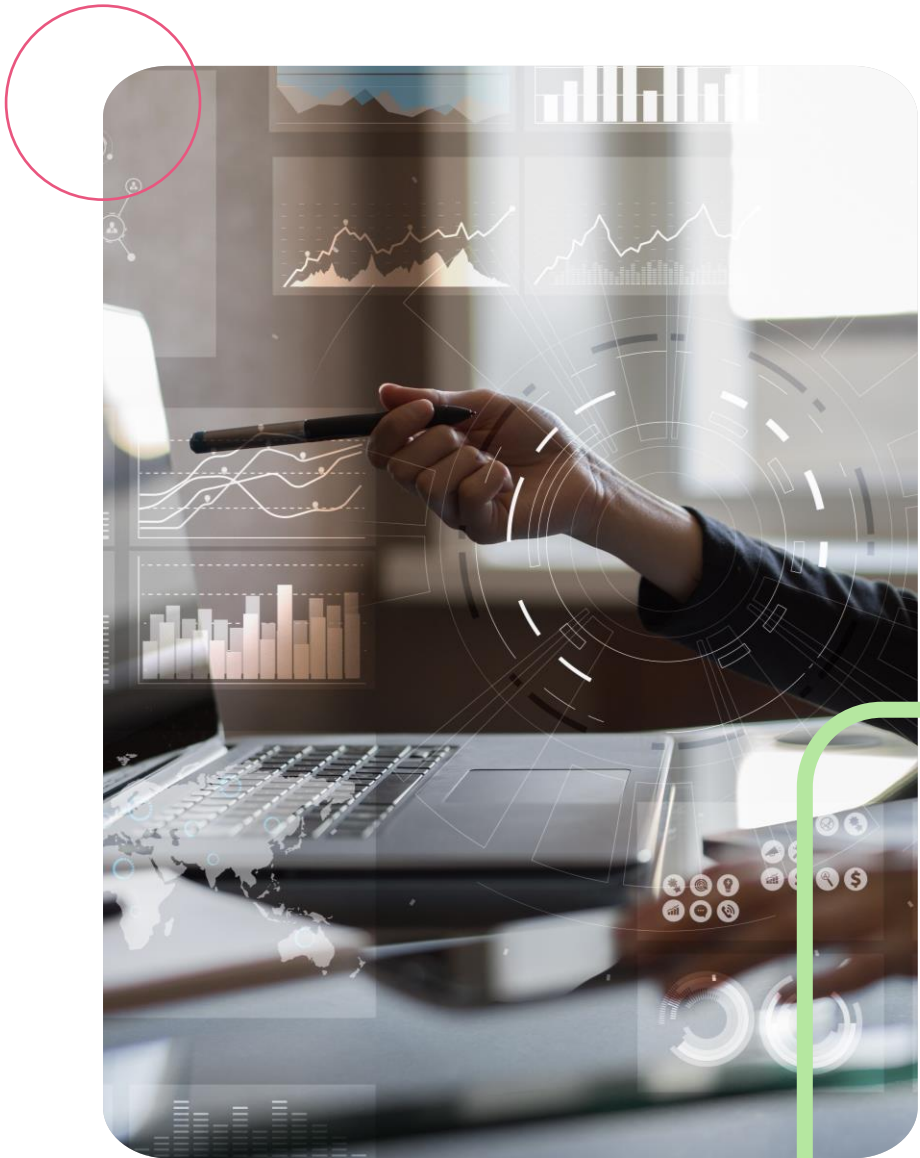




PROFESSIONAL DIPLOMA IN

Data Analytics

Lesson 8: Estimation



Small sample hypothesis testing (continued) <

Estimation <

Tying it all together <

Lesson Objectives

Lesson 8



Small sample hypothesis testing



Recap

Population standard deviation
unknown and small sample size
($n < 30$)

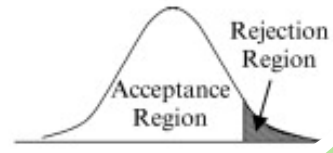
$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}}, d.f. = n - 1$$



Regions

Acceptance region

Provisionally
accept the null
hypothesis

One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X < \mu_0$	$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X \neq \mu_0$	$H_0 : \mu_X = \mu_0$ $H_1 : \mu_X > \mu_0$
		

Rejection region

Reject the null
hypothesis

Source: <https://www.sciencedirect.com/topics/mathematics/rejection-region>

Chocolate example: Step 1

State the hypothesis

H_0 : mean weight $\geq 100\text{g}$

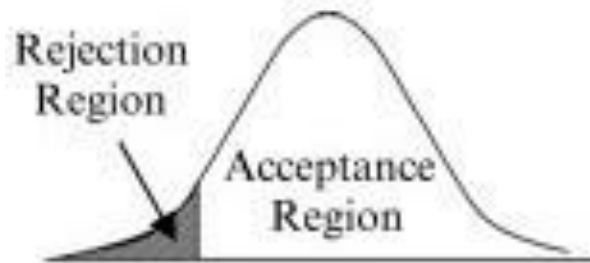
H_a : mean weight $< 100\text{g}$



Chocolate example: Step 2

95% confidence ($\alpha = 0,05$)

Critical value of -2,132 ($n = 5$)



Chocolate example: Critical t value

Degrees of freedom (ν)	Amount of area in one tail (α)					
	0.0005	0.001	0.005	0.010	0.025	0.050
1	636.6192	318.3088	63.65674	31.82052	12.70620	6.313752
2	31.59905	22.32712	9.924843	6.964557	4.302653	2.919986
3	12.92398	10.21453	5.840909	4.540703	3.182446	2.353363
4	8.610302	7.173182	4.604095	3.746947	2.776445	2.131847



Chocolate example

Step 3

$$\bar{x} = 99 \text{ g}, \mu = 100 \text{ g}, s = 1, n = 5$$

$$T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

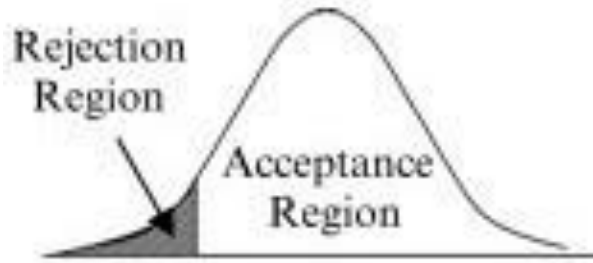
$$T = \frac{99 - 100}{1/\sqrt{5}}$$

$$T = -2,236$$



Chocolate example

Step 4



∴ test statistic in rejection region

∴ reject null hypothesis





Demo time! <<





Estimation

Estimation

- Point
 - Unbiased, consistent, efficient
- Interval
 - Confidence interval



Least squares method

- Line of best fit
- Line minimising sum of residuals

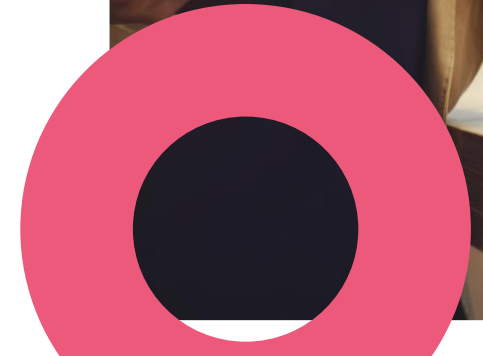




Why least squares?



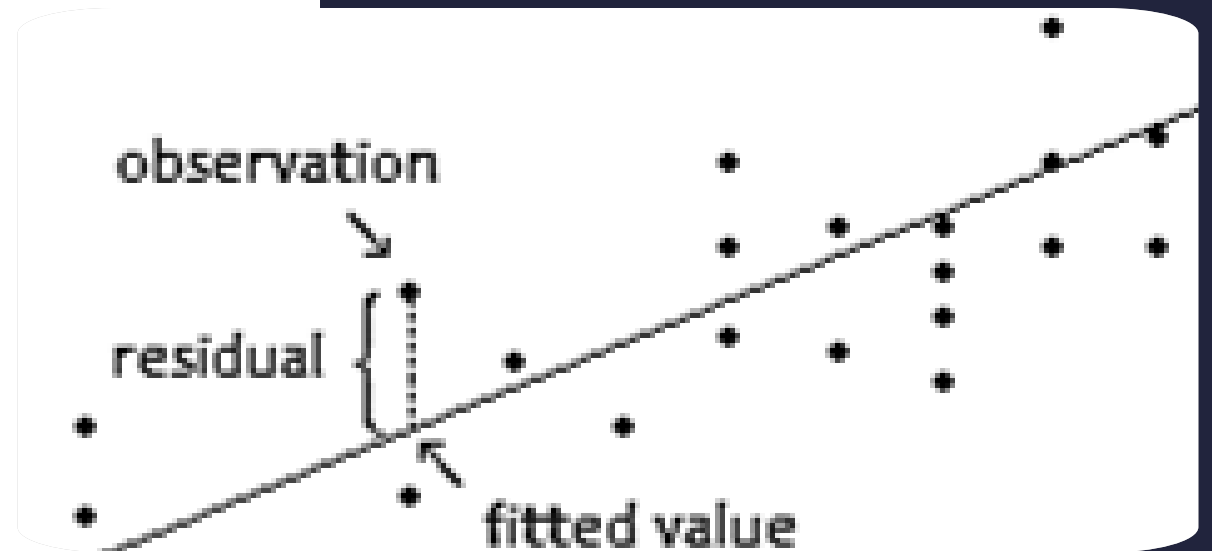
- Optimal fit to a distribution
- Fit a distribution to sample data that applies best to population data



Least squares calculation

$$\begin{aligned} \text{Residual} &= e_i = \text{observed} - \text{fitted value} \\ &= y - \hat{y} \end{aligned}$$

$$SSE = \sum e_i^2 = \sum_{i=1}^n (y - \hat{y})^2$$



Guitar example

Decibels measured at different
temperatures

Line of best fit helps us to
recognise relationships





**Tying it all
together**





Demo time! <<





Challenge >>

Recap concepts through
practise

#exploredata